Object Re-Identification with Locality Sensitive Hashing

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***Abstract*—Object re-identification on camera nodes depends on an explicit connection between the observed foreground region and the evidence used to retrieve it. This paper presents a classical pipeline that forms coherent object observations, extracts a masked 602-dimensional appearance descriptor, and retrieves candidate exemplars using random-hyperplane locality-sensitive hashing (LSH) followed by exact cosine verification. Core support, normalized size constraints, and temporal confirmation suppress isolated fragments before feature extraction. The descriptor combines regional color, multiscale texture, gradient structure, and channel statistics, while mutable hash tables retain references to bounded identity exemplars. Evaluation on a real parking-lot recording covers 9,484 frames over 316.13 seconds at 960×540 resolution. The run completes without frame-processing errors and achieves 72.55 frames/s of wall-clock replay throughput; human review establishes 69.23% precision among accepted match decisions. A separate synthetic-vector search benchmark measures a 4.77-fold query speedup over exhaustive cosine search at 10,000 vectors, retaining the exact nearest candidate for 98.33% of queries.**

***Index Terms*—Object Re-Identification, Foreground Segmentation, Locality-Sensitive Hashing.**

I. Introduction

Re-identification associates observations of a physical object when continuous tracking is unavailable, for example after the object leaves one camera and appears in another. The difficulty begins before retrieval: a foreground fragment, an oversized bounding box, or background contamination can change the appearance vector substantially. An efficient index cannot recover identity information that was lost while forming the observation. A useful system must therefore explain both what it bounds and how that bounded evidence becomes a searchable representation.

Recent object and vehicle Re-ID methods use transformer representations, self-supervised learning, and domain adaptation to improve discrimination under viewpoint and illumination changes [1]–[5]. These developments motivate complementary investigation of a transparent classical pipeline whose parameters, memory organization, and decisions can be inspected directly. Such a pipeline is useful for CPU-based prototypes and for studying the relationship between segmentation, representation, and candidate search without an additional training stage.

We present a stationary-camera pipeline that connects foreground support to temporal object formation, masked appearance embedding, binary signatures, and exact candidate verification. We use the term *object* for a coherent foreground component satisfying geometric and temporal conditions. This operational definition supports the vehicle-oriented prototype and whole-object demonstrations; semantic classification is a separate task. Likewise, a hash collision supplies a retrieval candidate rather than a unique physical identity.

The main contributions are:

1) An integrated observation-formation procedure combining a substantial foreground core, tighter relative dimensions, component-specific masks, and temporal confirmation. Connected fine detail is retained around a valid core, while isolated thin fragments are screened out.
2) A 602-value masked descriptor with auditable dimensional accounting and block-normalized color, texture, structure, and statistical evidence.
3) A random hyperplane index with radius-one probing, explicit exemplar references, and exact cosine verification, together with a derivation of its collision and candidate admission probabilities.

II. Related Work

A. Classical Segmentation and Object Formation

Adaptive Gaussian-mixture background subtraction estimates foreground in stationary-camera streams [6], [7]. Shadow separation and component geometry help turn this pixel support into useful observations. Our method adds an explicit thickness

test, independent width and height bounds, and foreground occupancy constraints before admitting an object crop.

B. Appearance Representations for Re-Identification

TransReID models appearance with patch-based transformers and uses camera/view embeddings to account for acquisition conditions [1]. Its jigsaw patch module rearranges patches to learn additional local features, complementing the global representation. GiT also combines local and global evidence, but does so through local correlation graphs that interact with transformer layers [2]. This interaction lets local details inform the global representation and global context guide local feature learning. SSBVER combines supervised identity learning with self-supervision through a teacher–student training scheme [3]. The additional training objective improves the learned representation without adding modules at deployment. CAViT adds contextual alignment for video objects [8]. Day–night adaptation [4], VehicleMAE pre-training [5], and modality-aware meta-learning [9] address appearance changes across acquisition conditions. Yi et al. survey the broader vehicle Re-ID landscape [10].

Our representation combines local binary patterns (LBP) [11], gradient histograms [12], and masked regional statistics. ORB supplies local correspondence evidence when enough keypoints are available [13]. This offers a deterministic descriptor without representation training.

C. Approximate Search and Tracking

SPANN combines an in-memory index of cluster centroids with disk-resident posting lists, reducing the memory needed for large vector galleries [14]. Its query-aware pruning limits which posting lists are read, addressing disk access as well as candidate search. SPFresh focuses on galleries that change over time and supports in-place vector updates [15]. Its incremental rebalancing splits partitions and reassigns nearby vectors, avoiding the cost of repeatedly rebuilding the entire index. Window-filtered ANN [16] and the NeurIPS ANN challenge [17] motivate evaluating eligibility, recall, and latency together. For temporal association, ByteTrack uses detection confidence [18]; our tracker instead assigns admissible foreground components using overlap, displacement, and scale change [19]. These roles separate within-view continuity from gallery retrieval.

D. Locality-Sensitive Hashing for Visual Retrieval

Random-hyperplane LSH maps angular similarity to a collision probability [20]. Multi-probe search visits neighboring buckets to increase candidate coverage without replicating the entire index [21]. More recent work revisits the bucket structure: DB-LSH constructs query-dependent buckets [22], and DET-LSH uses dynamic encoding trees to support efficient indexing and range queries [23]. These methods highlight the importance of bucket boundaries and index maintenance. We use fixed binary signatures with exhaustive one-bit probing, an auditable choice for mutable exemplar galleries. Full-vector cosine scoring verifies retrieved candidates, and an exhaustive scan supplies the numerical reference for query speed and nearest-neighbor retention.

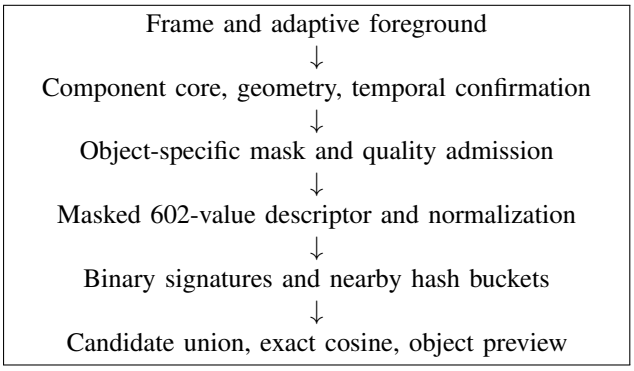


Fig. 1. Flowchart of the framework

III. Proposed Methodology

Overview of the Proposed Framework: Let a frame be $I_t \in \mathbb{R}^{H\times W\times 3}$. Each admitted observation contains a box $B = (x, y, w, h)$, component mask M, local track identifier ℓ, quality q, and descriptor $\mathbf{z}$. Figure 1 shows the processing sequence. No camera identifier, timestamp, or ground-truth identity is appended to the visual descriptor.

A. Foreground Support and Coherent Components

The default estimator is OpenCV MOG2 [6], [7], which updates a Gaussian mixture at each pixel to distinguish persistent background from foreground. Only mask value 255 is accepted as foreground; the shadow value 127 is kept separate. A four-second warmup initializes the background. A 3×3 elliptical opening and closing suppress small isolated noise, followed by filling enclosed holes of at most $0.001HW$ pixels.

For a connected component C, let $a_C = |C|$ and let $d(u, F^c)$ be the distance from pixel u to the cleaned foreground complement. A substantial component satisfies

$$\begin{aligned} a_C &\geq \max(9, \mathrm{round}(0.0008HW)), \\ \max_{u\in C} d(u, F^c) &\geq 0.015 \min(H, W). \end{aligned} \tag{1}$$

Here F is the cleaned foreground set, F^c its complement, u a pixel coordinate, and $|C|$ the number of pixels in component C; H and W are frame height and width. The distance transform uses a zero border and OpenCV's five-mask approximation. Crucially, Eq. (1) tests the presence of a core and then retains its entire connected component. Thin detail attached to a substantial foreground region is retained, while an isolated thin fragment fails admission. This preserves connected appearance evidence without broadly merging neighboring components.

B. Normalized Bounding-Box Admission

We define $r_y = \mathrm{clip}((y+h)/H, 0, 1)$. All of the following inequalities must hold:

$$\begin{aligned} 0.065 \le w/W \le 0.80, \\ 0.10 \le h/H \le 0.85, \\ 0.008(1 + 0.25 r_y) \le wh/(HW) \le 0.40, \\ a_C/(HW) \ge 0.004, \\ a_C/(wh) \ge 0.30, \qquad 0.4 \le w/h \le 3.5. \end{aligned} \tag{2}$$

In Eq. (2), (x, y) is the box's upper-left corner, w, h are its width and height, r_y is its normalized bottom position, and a_C is component area. The width, height, and upper-area bounds act independently. The minimum area increases slightly toward the image bottom, while foreground occupancy rejects mostly empty rectangles. At 960×540, the minimum integer width and height are 63 and 54 pixels, respectively, and the support floor is 2,074 pixels. These constants were empeerically devired. The lower size and support limits screen out small fragments; the upper limits reject oversized regions. The occupancy floor of 0.30 requires substantial foreground within a box, while the aspect-ratio interval admits both tall and wide views. The factor $1 + 0.25 r_y$ raises the minimum box area by up to 25% toward the image bottom to reflect the expected increase in apparent object size. Calibration on independent recordings is needed to establish how well these choices transfer to another camera placement.

C. Temporal Association and Observation Quality

A track predicts its center using smoothed velocity. For track i and component j, define center displacement δ_{ij} and scale change η_{ij} as

$$\begin{aligned} d_i &= \max(20, \sqrt{\widehat{w}_i^2 + \widehat{h}_i^2}), \\ \delta_{ij} &= \| \operatorname{ctr}(\widehat{B}_i) - \operatorname{ctr}(B_j) \|_2 / d_i, \\ \eta_{ij} &= \tfrac{1}{2} \left(\left| \log \tfrac{\widehat{w}_i}{w_j} \right| + \left| \log \tfrac{\widehat{h}_i}{h_j} \right| \right), \\ c_{ij} &= 0.45(1 - \mathrm{IoU}(\widehat{B}_i, B_j)) \\ &\quad + 0.40\delta_{ij} + 0.15\eta_{ij}. \end{aligned} \tag{3}$$

Here $\widehat{B}_i$ is track i's predicted box with dimensions $\widehat{w}_i, \widehat{h}_i$, B_j is candidate j's box with dimensions w_j, h_j, and ctr returns a box center. The scale d_i normalizes displacement; η_{ij} penalizes logarithmic size change and c_{ij} is the assignment cost. IoU is intersection area divided by union area. Size ratios outside $[0.45, 2.2]$ are rejected, as are pairs with IoU below 0.01 and $\delta_{ij} > 0.85$. A rectangular linear assignment produces one-to-one associations [19]. Invalid fragments are removed before assignment so they cannot shrink an established track.

Velocity is updated with weights 0.65 on the previous estimate and 0.35 on the newly measured velocity. Motion consistency s_t is an exponential moving average of normalized prediction agreement, with weight 0.3 on each new measurement. Confirmation requires six cumulative associated hits, $s_t \ge 0.4$, and valid geometry. Tracks expire after more than 14 missed updates. A merged component containing two established track centers is withheld so the tracks can coast separately.

The crop is masked by its own connected component. Quality admission uses

$$\begin{aligned} q_0 &= 0.20 q_s + 0.25 q_l + 0.20 q_f + 0.15 q_v + 0.20 s_t, \\ q_s &= \min(1, \min(h_c, w_c)/90), \\ q_l &= \min(1, \mathrm{Var}_M(\nabla^2 I)/180), \\ q_f &= \mathrm{clip}(1 - |\phi - 0.7|/0.7, 0, 1). \end{aligned} \tag{4}$$

In Eq. (4), q_0 is the initial quality score, h_c, w_c are crop dimensions, q_s scores size, and q_l scores sharpness using the masked variance of the image Laplacian $\nabla^2 I$. The fill score q_f uses foreground occupancy $\phi = |M|/(h_c w_c)$; q_v is the visible-box fraction and s_t is motion consistency. The constants 90 and 180 set size and sharpness saturation scales. Severe clipping, low support, or insufficient size multiplies quality by 0.3; flagged occlusion multiplies it by 0.45. The resulting penalized score is q; an observation is admitted at $q \ge 0.42$. These terms are engineering scores, not probabilities of a correct identity. The association weights give overlap and displacement greater influence than scale change; the 20-pixel floor prevents very small boxes from amplifying normalized displacement. Ratio and displacement gates exclude abrupt changes before assignment. The smoothing weights balance recent motion against the previous estimate, while six-hit confirmation and the 14-miss expiry delay admission and tolerate short gaps, respectively. In the quality score, the 90-pixel and 180-variance scales limit the influence of size and sharpness, and the fill term favors occupancy near 0.7. The quality weights, clipping and occlusion penalties, motion-consistency floor, and 0.42 admission threshold are fixed heuristic settings. They were held constant during evaluation; no controlled parameter-selection study is available to establish their optimality.

D. Masked Appearance Embedding

The crop is aspect-preserving letterboxed to 128×80 pixels. Image and mask resizing use area and nearest-neighbor interpolation, respectively. Masked statistics exclude padding; texture and gradients use a 7×7 eroded interior, reverting to the original support if fewer than ten pixels remain. The 602 descriptor values comprise 180 color values (156 regional histogram bins and 24 Lab cell moments), 54 texture values, 332 structure values, and 36 statistical values. This fixed decomposition makes the representation compact and reproducible.

1) Color and joint statistics: The whole crop and its upper and lower halves each contribute HSV histograms with 12/8/8 bins and Lab histograms with 8/8/8 bins. A 2×2 Lab partition contributes means and standard deviations. Each histogram count is divided by the total masked count and square-rooted, reducing domination by highly populated bins. The statistical block uses normalized HSV/Lab values: six channel means, standard deviations, and interquartile ranges contribute 18 values, and the 15 off-diagonal sample covariances describe inter-channel dependence. Mask occupancy, normalized aspect

ratio, and mean gradient magnitude supply the final three values.

2) Texture and structure: Eight circular neighbors at radius $R \in \{1,2,3\}$ define bits $b_p = \mathbf{1}[g_p \geq g_c]$. With circular transition count $U = \sum_{p=0}^{7} |b_p - b_{(p+1) \bmod 8}|$, uniform rotation-invariant LBP [11] is

$$\mathrm{LBP}^{\mathrm{riu2}}_{8,R} = \begin{cases} \sum_{p=0}^{7} b_p, & U \leq 2, \\ 9, & U > 2. \end{cases} \tag{5}$$

Here g_c is center-pixel grayscale intensity, g_p is neighbor p's interpolated intensity, $\mathbf{1}[\cdot]$ is one when its condition holds and zero otherwise, and U counts circular bit transitions. The radius R is measured in canonical-crop pixels; codes 0–8 count set bits in uniform patterns, while code 9 collects the remaining patterns. Each radius supplies ten LBP bins and eight log-variance bins. Sobel gradients form nine unsigned-orientation bins in each of 4×8 cells, inspired by HOG [12]. Each cell accumulates gradient magnitudes by unsigned orientation over the masked interior, then normalizes its nine-bin histogram, contributing 288 values.

Nine Canny edge-orientation bins and edge density add ten values. A 6×4 silhouette and its four row and six column means add coarse occupancy structure. The local LBP rotation property does not imply full viewpoint invariance of the combined descriptor.

3) Block normalization and cosine scoring: Let $\mathcal{U}(\mathbf{f}) = \mathbf{f} / \max(\|\mathbf{f}\|_2, 10^{-12})$. For blocks C, T, G, S and weights $\boldsymbol{\alpha} = (1, 0.65, 0.70, 0.40)$,

$$\begin{aligned} \mathbf{z} &= \mathcal{U}\left(\Big\Vert_{b \in \{C,T,G,S\}} \alpha_b \mathcal{U}(\mathbf{f}_b) \right), \\ \mathbf{z} &\in \mathbb{R}^{602}, \qquad s(\mathbf{z}, \mathbf{z}') = \mathbf{z}^\top \mathbf{z}'. \end{aligned} \tag{6}$$

Here $\mathbf{f}_b$ is feature block b: C, T, G, S denote color, texture, structure, and statistics. The operator $\mathcal{U}$ performs L2 normalization with a small numerical floor, α_b is its block weight, and $\Vert$ denotes concatenation. The final unit vector $\mathbf{z}$ is compared with another observation $\mathbf{z}'$ by dot-product cosine s. Independent normalization prevents block length alone from determining influence; the chosen weights contribute quadratically to cosine when every block is nonzero. A descriptor-profile checksum records the feature version and weights for compatibility.

E. Hash Generation and Table Organization

For table l and bit j, draw independent $\mathbf{r}_{lj} \sim \mathcal{N}(\mathbf{0}, I_{602})$ with a recorded random seed. The signature is

$$\begin{aligned} h_{lj}(\mathbf{z}) &= \mathbf{1}[\mathbf{r}_{lj}^\top \mathbf{z} \geq 0], \\ H_l(\mathbf{z}) &= \sum_{j=0}^{b-1} 2^j h_{lj}(\mathbf{z}). \end{aligned} \tag{7}$$

Here l indexes a table, j a bit, $\mathbf{r}_{lj}$ is a Gaussian projection-plane normal, and I_{602} is the identity covariance matrix. The binary output h_{lj} records the side of the plane; H_l packs b such bits into an integer. The default $L = 8$ tables and $b = 12$ bits produce eight integers in $[0, 4095]$. Planes and descriptors use float32, and bit zero is least significant. Similar vectors tend to share signatures, but many distinct objects can collide.

For a fixed unit-vector pair with angle θ, a single random-hyperplane bit agrees with probability $p = 1 - \theta/\pi$ [20]. Independence gives

$$\begin{aligned} \theta &= \arccos(\mathrm{clip}(\mathbf{z}^\top \mathbf{z}', -1, 1)), \\ q_0 &= p^b, \qquad q_1 = p^b + b(1-p)p^{b-1}, \\ P_r &= 1 - (1 - q_r)^L, \qquad r \in \{0, 1\}. \end{aligned} \tag{8}$$

Here θ is the pair's angle, p is one-bit agreement probability, q_0 is exact-signature agreement, and q_1 allows zero or one differing bit in a table. The radius r selects exact-only or one-bit probing, and P_r is admission by at least one of L independent tables. Clipping keeps the inverse cosine within its numerical domain. These probabilities describe random index construction for a fixed pair, not the probability that two observations show the same physical object.

Each dictionary maps an integer signature to a set of exemplar keys:

$$\begin{aligned} T_l[a] &= \{k : H_l(\mathbf{z}_k) = a\}, \\ R[k] &= (H_1(\mathbf{z}_k), \ldots, H_L(\mathbf{z}_k)). \end{aligned} \tag{9}$$

Here k identifies an exemplar, a is an integer bucket signature, and $T_l[a]$ is the set of exemplar keys stored in that bucket of table l. The reverse map R records each key's signatures and supports removal without scanning all buckets. Updating a key first removes its old references, then inserts its new signature into each table. Empty buckets are deleted. Full descriptors remain in the gallery for exact scoring; hash tables store references to them.

The live gallery retains up to eight exemplars per identity. A near duplicate with cosine at least 0.985 replaces an older exemplar when quality improves. Retention starts with the highest-quality item, then balances quality (weight 0.55) against cosine diversity from already selected exemplars (weight 0.45). Removing an exemplar also removes its bucket references, keeping the descriptor store and hash tables consistent.

F. Multi-Probe Retrieval and Exact Verification

For an eligible-gallery set E_q, define the per-table probe signatures

$$\begin{aligned} \mathcal{A}_l(q) &= \{H_l(\mathbf{z}_q)\} \\ &\quad \cup \{H_l(\mathbf{z}_q) \oplus 2^j : 0 \leq j < b\}, \\ \mathcal{K}_q &= E_q \cap \bigcup_{l=1}^{L} \bigcup_{a \in \mathcal{A}_l(q)} T_l[a], \\ \widehat{k} &= \arg\max_{k \in \mathcal{K}_q} \mathbf{z}_q^\top \mathbf{z}_k. \end{aligned} \tag{10}$$

In Eq. (10), q identifies the query, E_q is the set of eligible gallery keys, $\mathcal{A}_l(q)$ lists the signatures probed in table l, and $\mathcal{K}_q$ is their deduplicated candidate union. XOR ($\oplus$) flips bit j; $\widehat{k}$ denotes the highest-cosine retained exemplar. Empty candidate sets provide no indexed match.

Equation (10) describes the search primitive. In the application, retrieved keys nominate identities, and exact scoring

Algorithm 1 Visual observation and indexed retrieval

Require: Frame I_t, track state, gallery, tables T_l
Estimate foreground and separate shadow support
Clean components; apply core and size tests
Associate admissible components with track predictions
for each currently confirmed track due for observation **do**
  Form component-specific crop and mask
  Compute quality using Eq. (4)
  if $q \geq 0.42$ **then**
    Extract normalized descriptor (Eq. (6))
    Compute signatures and collect $\mathcal{K}_q$
    Rank candidates by exact cosine (Eq. (10))
    Retain evidence and the observation preview
    Update retained exemplar references when admitted
  end if
end for

selects the strongest retained exemplar of each nominated identity. An already bound local track checks its existing identity, while identities occupied by other active tracks are excluded from a new assignment. Thus repeated confirmations and new associations can both generate a match decision.

For the recorded run, available ORB evidence refines global visual cosine with weights 0.1 and 0.9, respectively. The application combines visual, temporal, and spatial scores with relative weights 0.78, 0.07, and 0.05, normalized over available channels. Context scores use elapsed time, position, and scale outside the appearance vector. No supported plate reading occurs in the archived observations. A visual floor of 0.60 and fused threshold of 0.82 admit `MATCH`; scores from 0.68 below 0.82 are `UNCERTAIN`. A gap below 0.035 between the two best viable identities also causes uncertainty. Human review evaluates these complete application decisions, rather than treating every hash collision as a successful Re-ID.

G. Hyperparameter Selection

The parking-lot replay uses the recorded operational configuration: MOG2 history 500 and variance threshold 24; a four-second warmup; the geometric constraints in Eq. (2); six-hit confirmation; quality floor 0.42; and observations spaced by at least 1.5 seconds of video time per confirmed track. The index uses eight 12-bit tables, radius-one probing, and seed 17, with at most eight exemplars per identity. These settings remain fixed throughout the recording, and reviewer labels do not update descriptors, thresholds, or subsequent predictions.

H. Computational Complexity

With fixed background-model and morphology settings, foreground processing scales with the HW image pixels. For n tracks and m components, forming association costs takes $O(nm)$; dense assignment has a cubic worst-case bound in the larger dimension. Canonical feature extraction has fixed image support and descriptor dimension $D = 602$.

Hash projection costs $O(LbD)$ per observation. Radius-one probing visits $L(b+1)$ buckets, or 104 by default. If V_q is the number of bucket references visited, candidate collection and scoring cost

$$T_q = O\big(LbD + V_q + |\mathcal{K}_q|D\big). \tag{11}$$

Here T_q is query work, D is descriptor dimension, V_q counts bucket references visited (including repeated keys), and $|\mathcal{K}_q|$ is the number of unique eligible candidates. Bounded top-k selection, where k is the number of requested neighbors, adds candidate-dependent selection work. Dense buckets can make V_q large; no unconditional sublinear query bound follows. Exhaustive scoring costs $O(ND)$ but benefits from contiguous matrix operations. For a gallery of N exemplars, storage consists of ND vector values, LbD plane values, and LN key memberships, with Python dictionary/set overhead measured separately. This analysis predicts an empirical break-even region rather than universal LSH superiority.

IV. Experimental Results

A. Experimental Setup

1) Parking-lot recording and deployment: The field evaluation uses a real parking-lot MP4 containing 9,484 frames at 30 frames/s, spanning 316.13 seconds (5 minutes 16 seconds). Frames are processed at 960×540 resolution. The first 120 frames initialize the background model; the remaining 9,364 frames enter operational evaluation. File replay decodes every frame in order, uses frame index divided by frame rate as video time, and runs as fast as processing permits. It exercises the application pipeline with actual recorded imagery.

The LAN test deployment included a Latitude laptop node equipped with an Intel Core i5-2520M. The video replay and LSH timing measurements were collected on a Windows host. The LSH timing suite pins OpenCV and BLAS to one thread [24]. Reported throughput and query timings belong to those local runs; the i5 node describes the LAN deployment and was not benchmarked in these experiments. The research-suite environment records Python 3.14.3, NumPy 2.5.3, SciPy 1.18.1, and OpenCV 4.14.0.94.

2) Human review and metric scope: The reviewer starts replay in the browser. Each `MATCH` decision contributes a frozen pair consisting of the current object crop and the selected earlier exemplar. The reviewer approves a pair if both show the same physical object and rejects it otherwise. The queue excludes `NO_MATCH` and `UNCERTAIN`. Review labels are stored separately from the application's predictions; the final report is exported after both video completion and completion of every queued review.

Let $\mathcal{M}$ be the set of accepted match decisions, and let $y_i = 1$ indicate human approval of pair i, with $y_i = 0$ indicating rejection. Accepted-match precision is

$$\begin{aligned} n_{\mathrm{a}} &= \sum_{i \in \mathcal{M}} y_i, \qquad n_{\mathrm{r}} = |\mathcal{M}| - n_{\mathrm{a}}, \\ P_{\mathrm{match}} &= \frac{n_{\mathrm{a}}}{n_{\mathrm{a}} + n_{\mathrm{r}}}. \end{aligned} \tag{12}$$

Here n_{a} and n_{r} count approved and rejected pairs. Since every reviewed prediction is positive, agreement on this selected set

TABLE I
PARKING-LOT PROCESSING AND HUMAN-REVIEW RESULTS FOR THE 316.13-SECOND RECORDING.

Measurement	Result
Input resolution / frame rate	960×540 / 30 FPS
Warmup / evaluated frames	120 / 9,364
Candidate-box instances	5,160
Confirmed-track frame instances	3,946
Frame-processing errors	0
Observation decisions	214
Matches / uncertain outcomes	13 / 179
Non-matches / quality skips	16 / 6
Reviewed / queued matches	13 / 13
Approved / rejected pairs	9 / 4
Accepted-match precision	69.23%
Replay wall time	130.72 s
Wall-clock replay throughput	72.55 FPS
Mean frame-processing time	11.32 ms
Processing-only throughput	88.30 FPS

equals its precision. Unreviewed non-matches do not establish true-negative or false-negative counts; full recall, F1, rank-1, mAP, and detection/mask accuracy therefore are not inferred from this protocol. Repeated confirmations of a tracked object are valid review events but are not independent physical identities or cross-camera transitions.

Percentages are computed from the individual reviewer labels and checked against the corresponding counts. Each judgment is linked to the observation frame and the two images used for the decision.

B. Quantitative Parking-Lot Results

Table I summarizes the complete recording. Processing produces 5,160 candidate-box instances and 3,946 confirmed-track frame instances, with no frame-processing errors. These are accumulated processing counts; one object can contribute on several frames. The 214 observation decisions are divided into accepted matches, uncertain outcomes, non-matches, and quality skips in Table I. The final gallery contains 16 system-assigned identities, which are not ground-truth vehicle counts.

Human review yields $P_{\mathrm{match}} = 69.23\%$ among accepted decisions, with the supporting counts and complete review coverage reported in Table I. The limited review sample measures performance on this recording; it does not establish general Re-ID accuracy. Most decisions remain uncertain under the fixed acceptance policy, and those outcomes are outside the precision denominator.

C. LSH Query Speed and Scaling

The separate search experiment uses synthetic normalized 602-dimensional vectors to isolate querying speed from image acquisition and descriptor extraction. Gaussian identity centers receive independent noise with standard deviation 0.018, with five gallery vectors per center. Gallery sizes are 100, 1,000, 5,000, and 10,000, each with 100 independently generated queries. Data seed 2026 is fixed; plane seeds 17, 29, and 43 repeat index construction. These generated vectors are used only for the index-scaling experiment, not for the parking-lot precision result.

TABLE II
QUERY LATENCY AND NEAREST-NEIGHBOR RETENTION ON SYNTHETIC VECTORS, AVERAGED OVER THREE INDEX SEEDS.

Gallery	Scan (ms)	LSH (ms)	Speedup	Top-1 (%)
100	0.0095	0.0441	$0.22\times$	96.67
1,000	0.0356	0.0510	$0.70\times$	96.33
5,000	0.2976	0.1033	$2.88\times$	96.67
10,000	1.0805	0.2267	$4.77\times$	98.33

The numerical baseline is vectorized exhaustive cosine search on the same vectors and eligibility conditions. Five untimed query warmups precede five timed repetitions per query, with randomized interleaving of methods. Timing covers projection, probing, candidate collection, eligibility filtering, exact scoring, and top-10 selection. For each seed, the query latency is the median of per-query repetition medians; Table II averages those seed-level values and their paired scan/LSH ratios.

Candidate fraction is the number of unique scored candidates divided by eligible gallery size, averaged across queries. Exact-top-1 retention is the fraction of queries whose exhaustive nearest descriptor survives candidate generation. This fidelity check accompanies latency so that reduced work is not mistaken for equally complete retrieval. At 10,000 vectors, the index scores 2.63% of the gallery, retains the exact nearest candidate for 98.33% of queries, and achieves a $4.77\times$ speedup. Speedup ranges from 4.74 to 4.79 across the three seeds. At 5,000 vectors it reaches $2.88\times$; at the two smaller sizes, the vectorized scan is faster. The measured advantage therefore applies to the larger tested galleries and their angular distribution, not automatically to the small online gallery of the video run.

D. Qualitative Analysis

The review interface presents the exact observation and reference crops used by each accepted decision. Figure 2 shows one approved and one rejected pair from the archived parking-lot run. The approved pair shows consistent vehicle appearance. The rejected pair contains partial foreground over roadway rather than clear whole-vehicle crops, illustrating that geometric and quality gates can still admit ambiguous evidence. Displaying both stored images makes the reviewer's judgment assessable in a way that an identity tag alone cannot.

Component support, bounded crops, and exact candidate previews provide an chain from pixels to a proposed association. The preserved evidence makes subsequent analysis of lighting, partial foreground, and competing vehicles possible without reconstructing the live gallery.

E. Runtime Performance

The 316.13-second recording completes in 130.72 seconds of time, corresponding to 72.55 decoded frames/s and approximately $2.42\times$ the source playback rate on the replay host. Mean successful frame-processing time is 11.32 ms, corresponding



(a) Human-approved match, frame 646



(b) Human-rejected match, frame 1442

Fig. 2. First approved and first rejected pairs in review order, extracted unchanged from the stored parking-lot review images.

to 88.30 processing frames/s. The latter includes warmup, JPEG encoding, and per-frame application work but excludes capture wait; wall throughput uses the complete replay interval. Reviewer activity can overlap processing, and time spent after video completion is excluded from replay throughput.

The results show sufficient processing capacity for this recording's 30-frame/s timeline on the measured host. LSH query speed is reported separately because it times a retrieval primitive rather than the entire frame pipeline.

F. Discussion

The field run establishes that the complete application can decode a parking lot sequence, maintain coherent observations, propose associations, and preserve enough evidence for human verification. Its 69.23% match precision is an explicit operating point measurement. The uncertain outcomes and four rejected pairs point to threshold calibration and appearance discrimination as potential next improvements. Wider camera coverage and independently reviewed non accepted outcomes would extend this evaluation to missed associations and cross-camera identity retrieval.

The methodological distinctions have been diccussed: geometric and temporal support establish an operational object region; the descriptor summarizes its masked appearance; LSH narrows the candidate set; exact scoring and the acceptance policy produce a decision.

V. Conclusion and Future Work

We present a classical object Re-ID system linking foreground support, constrained bounding boxes, masked 602-dimensional descriptors, mutable random-hyperplane tables, and exact candidate verification. A real parking-lot evaluation covers 9,484 frames over 316.13 seconds, with no frame-processing errors, 72.55 FPS wall-clock replay throughput, and 69.23% precision among human-reviewed accepted matches. A separate synthetic-vector query benchmark demonstrates a $4.77\times$ LSH speedup at 10,000 vectors with 98.33% exact-neighbor retention. Together, these results establish a working visual pipeline and a measured search-scaling benefit.

Future work will extend human annotation to non-accepted candidates, calibrate acceptance on held-out recordings, and evaluate shared query/gallery identities across multiple cameras. Photometric normalization and a measured choice between exhaustive and indexed search can improve robustness and small-gallery efficiency. The preserved object pairs, explicit feature construction, and archived run configuration provide a practical foundation for these extensions.

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